Math 133 Written Homework 8

Applied Optimization

Due Thursday, November 12, 2026, at 11:59 p.m.

Show your work and use derivatives to justify absolute optima on the stated or feasible domain, including relevant endpoints. For applications, define your variables and state the function you optimize and its feasible domain. Treat quantities as continuous. Keep exact values until the final step; a calculator may be used for approximations. Give business answers with units, rounding prices and profit to the nearest cent and sales quantities to the nearest whole item when necessary.

List collaborators, time spent, and any use of generative AI with your submission. Your submitted work must be your own. Late work will not be accepted.

1. A manufacturer models its weekly profit by

$$P(x) = -2x^2 + 32x - 60, \qquad 0 \le x \le 12,$$

   where x is production in thousands of units and $P(x)$ is profit in thousands of dollars. How many units should it produce to maximize weekly profit, and what is that maximum profit?

2. A booth sells insulated water bottles. Its daily price-demand model is

$$p(q) = 28 - 0.04q, \qquad 0 \le q \le 700,$$

   where q is the number of bottles sold and $p(q)$ is the price in dollars per bottle. Daily costs are \$180 plus \$8 per bottle sold. Find the price that maximizes daily profit, the corresponding sales quantity, and the maximum daily profit.

3. At a daily rental price of \$25, a bike rental shop rents all 80 of its bikes. For each \$1 increase in price, two fewer bikes are rented per day. Daily operating costs are \$120 plus \$5 for each bike rented. The shop is considering prices of at least \$25. What rental price maximizes daily profit? Give the number of bikes rented and the maximum daily profit.

4. A company models weekly demand for a refill product by

$$p(x) = 8 - 2\ln x, \qquad 1 \le x \le 12,$$

where x is sales in thousands of units and $p(x)$ is the price in dollars per unit. Each unit costs \$2 to produce, and fixed operating costs are \$4,000 per week. Determine the price that maximizes weekly profit, the corresponding sales quantity, and the maximum weekly profit.

5. Find the absolute maximum and absolute minimum values of

$$f(x) = (x^2 + 1)e^{-x/2}$$

on $[0, 5]$. State every x-value where each is attained.

6. Find the absolute maximum and absolute minimum values of

$$g(x) = \frac{x^2 + 4x + 8}{x + 2}$$

on $[-1, 2]$. State every x-value where each is attained.